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3031ICT: Applied Data Mining
3804ICT: Data Mining

Lecture 4: Mining Frequent Patterns I



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Mining Frequent Patterns I

- Basic Concepts
- Frequent Itemset Mining Method – Apriori Algorithm
- Which Patterns Are Interesting?—Pattern Evaluation Methods
- Summary

What Is Frequent Pattern Analysis?

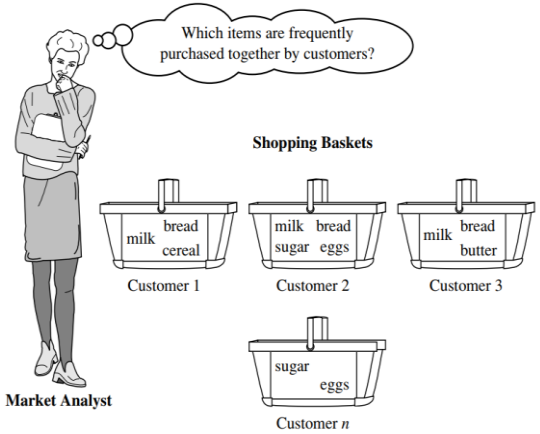
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- **Frequent pattern:** a pattern (**a set of items, subsequences, substructures**, etc.) that occurs frequently in a data set
  - **A set of items** (itemsets)
  - **A subsequence** (a frequent sequential pattern)
  - **A substructure** can refer to different structural forms,
- **Motivation**: Finding inherent regularities in data
  - What products are often purchased together?— Beer and diapers?!
  - What are the subsequent purchases after buying a PC?
  - What kinds of DNA are sensitive to this new drug?
  - Can we automatically classify web documents?
- **Applications**
  - Basket data analysis, cross-marketing, catalog design, sale campaign analysis, Web log (click stream) analysis, and DNA sequence analysis.

A Motivating Example of Market Basket Analysis

Managers' Question: ***"Which groups or sets of items are customers likely to purchase on a given trip to the store?"***

- Manager can then use the results to plan marketing or advertising strategies
  - *Layout 1* (**close**): items that are frequently purchased together can be placed in proximity to further encourage the combined sale of such items.
  - *Layout 2* (**far** & **opposite**): placing hardware and software at opposite ends of the store may force customers who purchase such items to pick up other items along the way.
- Market basket analysis can also help retailers plan which items to put on sale at reduced prices.



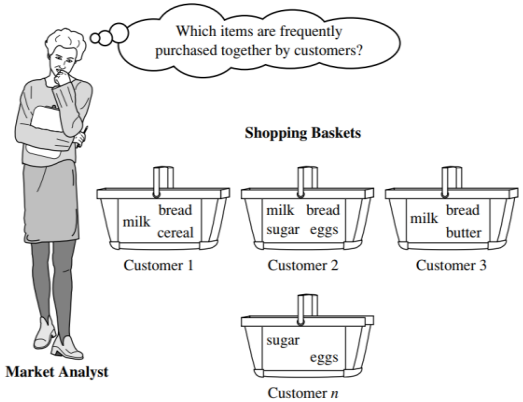
Market Basket Analysis process analyzes customer buying habits by finding associations between the different items that customers place in their "shopping baskets".

A Motivating Example of Market Basket Analysis

Patterns can be represented in the form of **association rules**.

$$computer \Rightarrow antivirus_software\ [support = 2\%, confidence = 60\%].$$

- **Support** and **confidence** are two measures of rule interestingness.
  - A ***support of 2%*** means that *2% of all the transactions under analysis show that computer and antivirus software are purchased together*.
  - A ***confidence of 60%*** means that *60% of the customers who purchased a computer also bought the software*.
- Association rules are considered interesting if they satisfy both a **minimum support threshold** and a **minimum confidence threshold**.
- Thresholds can be set by **users** or **domain experts**.



Basic Concepts: Support

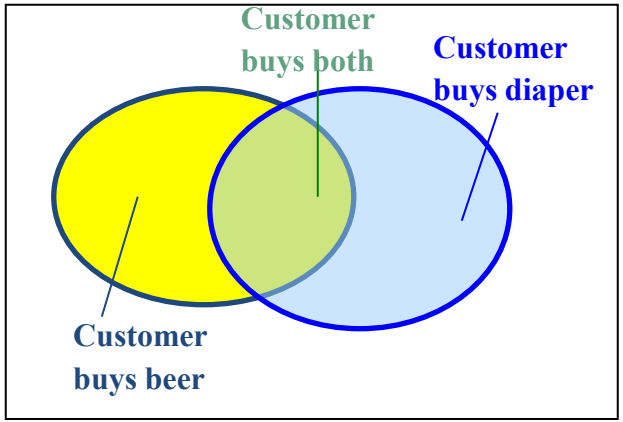
- **Itemset**
  - A collection of one or more items
    - Example: {Milk, Bread, Diaper}
  - **k-itemset**
    - An itemset that contains k items

      {Milk, Bread, Diaper} is a 3-itemset

      {Bread, Milk} is a 2-itemset
- **Support Count (Frequency, Count)**
  - Occurrence Frequency of an itemset
  - E.g. σ({Milk, Bread,Diaper}) = 2, What about σ({Bread, Milk}) = ?
- **Support**
  - Fraction of transactions that contain an itemset
  - E.g. s({Milk, Bread, Diaper}) = 2/5 s({Milk, Bread}) = ?
- **Frequent Itemset**
  - An itemset whose support is greater than or equal to a threshold *min_sup.*

    The set of frequent k-itemsets is denoted by F_k.

TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke



Basic Concepts: Association Rule

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Association Rule

An implication expression of the form $X \rightarrow Y$, where X and Y are itemsets

Example:

{Milk, Diaper} →{Beer}

Antecedent → Consequent

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Rule Evaluation Metrics

Support (s) Usefulness

- Fraction of transactions that contain both X and Y

Confidence (c) Certainty

- Measures how often items in Y appear in transactions that contain X

$$c = \frac{\sigma(X \cup Y)}{\sigma(X)}$$

Example:

$$\text{Rule} : \{\text{Milk, Diaper}\} \rightarrow \text{Beer}$$

$$s = \frac{\sigma(\{\text{Milk, Diaper, Beer}\})}{T} = \frac{2}{5} = 0.4$$

$$c = \frac{\sigma(\{\text{Milk, Diaper, Beer}\})}{\sigma(\{\text{Milk, Diaper}\})} = \frac{2}{3} = 0.67$$

A support of 0.4 for the rule above ({Milk, Diaper} -> Beer) means that 40% of all the transactions show that purchases of Milk and Diaper are with Beer.

A confidence of 0.67 means that 67% of the customers who purchased both Milk and Diaper also bought the Beer.

Basic Concepts: Association Rule

Association Rule

An implication expression of the form $X \rightarrow Y$, where X and Y are itemsets

Example:

{Milk, Diaper} →{Beer}

Antecedent → Consequent

TID	*Items*
1	Bread, Milk
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3	Milk, Diaper, Beer, Coke
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5	Bread, Milk, Diaper, Coke

Two thresholds (specified by users or experts):

- **Min_sup**: minimum support threshold
- **Min_conf**: minimum confidence threshold

Rules that satisfy **both** a minimum support threshold (min_sup) and a minimum confidence threshold (min_conf) are called **strong**.

$$confidence(A \Rightarrow B) = P(B|A) = \frac{support(A \cup B)}{support(A)} = \frac{support_count(A \cup B)}{support_count(A)}.$$

An Example Illustrates Basic Concepts

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- **Itemset**: there are eight items. Any combination can be an itemset
- **K-itemset**: consider 2-itemsets, e.g. {b,c}
- **Frequency / support count / count / absolute support**:
  - Count({b,c})=?
- **Support / relative support:**
  - support of {b,c}?
- **Min_sup**: 80% (0.8)
- **Frequent itemset**:
  - is itemset {b,c} frequent or not?
  - And itemset {a,c}?
- **Association Rule**: {b} -> {c} and {b,c} -> {h}
- **Confidence**: conf ({b} -> {c}) and conf ({b,c} -> {h})?
- **Min_conf**: 80%
- **Strong Rule**:
  - Rule {b} -> {c} strong? Rule {b,c} -> {h} strong?

TID	Items
1	{a, b, c}
2	{b, c, f, g}
3	{a, b, c, g, h}
4	{b, c}
5	{c, g, i}
6	{a, b, c, d, h}
7	{b, c, g, i, h}
8	{a, c, h}
9	{b, c, i}
10	{b, c, h}

Association Rule Mining Task

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Thus, the general approach to mining association rules is:

1. Frequent Itemset Generation
   - Generate all itemsets that satisfies the min_sup threshold.
   - These itemsets are called **frequent itemsets.**
2. Rule Generation
   - Generate high confidence rules from each frequent itemset, these rules must satisfy minimum support and minimum confidence, where each rule is a binary partitioning of a frequent itemset.
   - These rules are called **strong rules.**

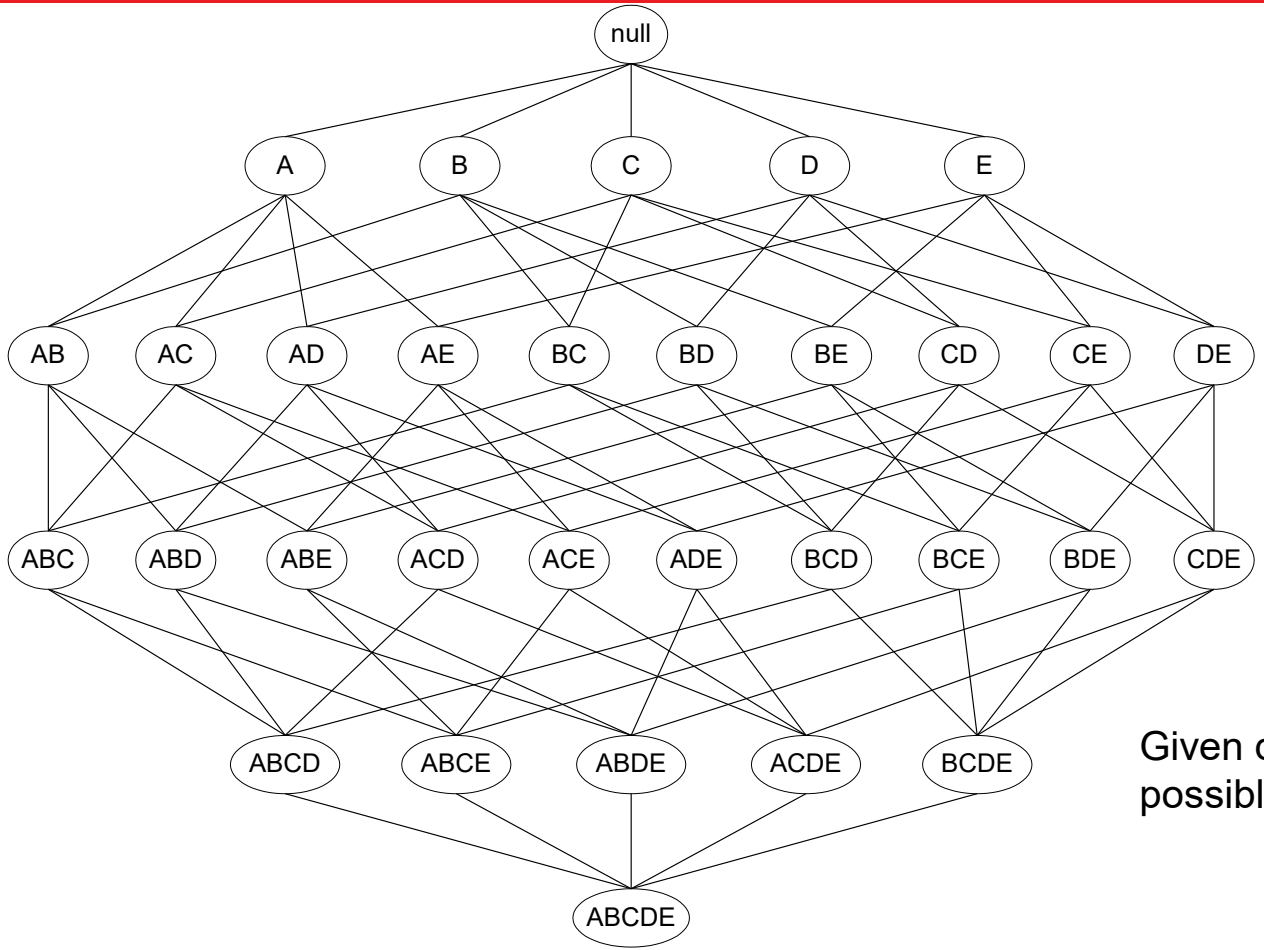
Frequent itemset generation is still computationally expensive!

Freq. Itemsets Generation is Pricy

- Step 1 often generates a huge number of itemsets satisfying the minimum support (*min_sup*) threshold, especially when *min-sup* is set low. This is because if an itemset is frequent, each of its subsets is frequent as well.
  - E.g. {a,b,c,d} is frequent, then {a}, {b}, {c}, {d}, {a,b},…,{b,c,d} are frequent as well.

- A long itemset will contain a combinatorial number of shorter, frequent sub-itemsets.
  - A long pattern contains a combinatorial number of sub-patterns, e.g., $\{a_1, \ldots, a_{100}\}$ contains $\binom{100}{1} + \binom{100}{2} + \cdots + \binom{100}{100} = 2^{100} - 1 \approx 1.27 \times 10^{30}$. sub-patterns!

Frequent Itemset Generation



Given d items, there are 2^d possible candidate itemsets

Apriori Principle

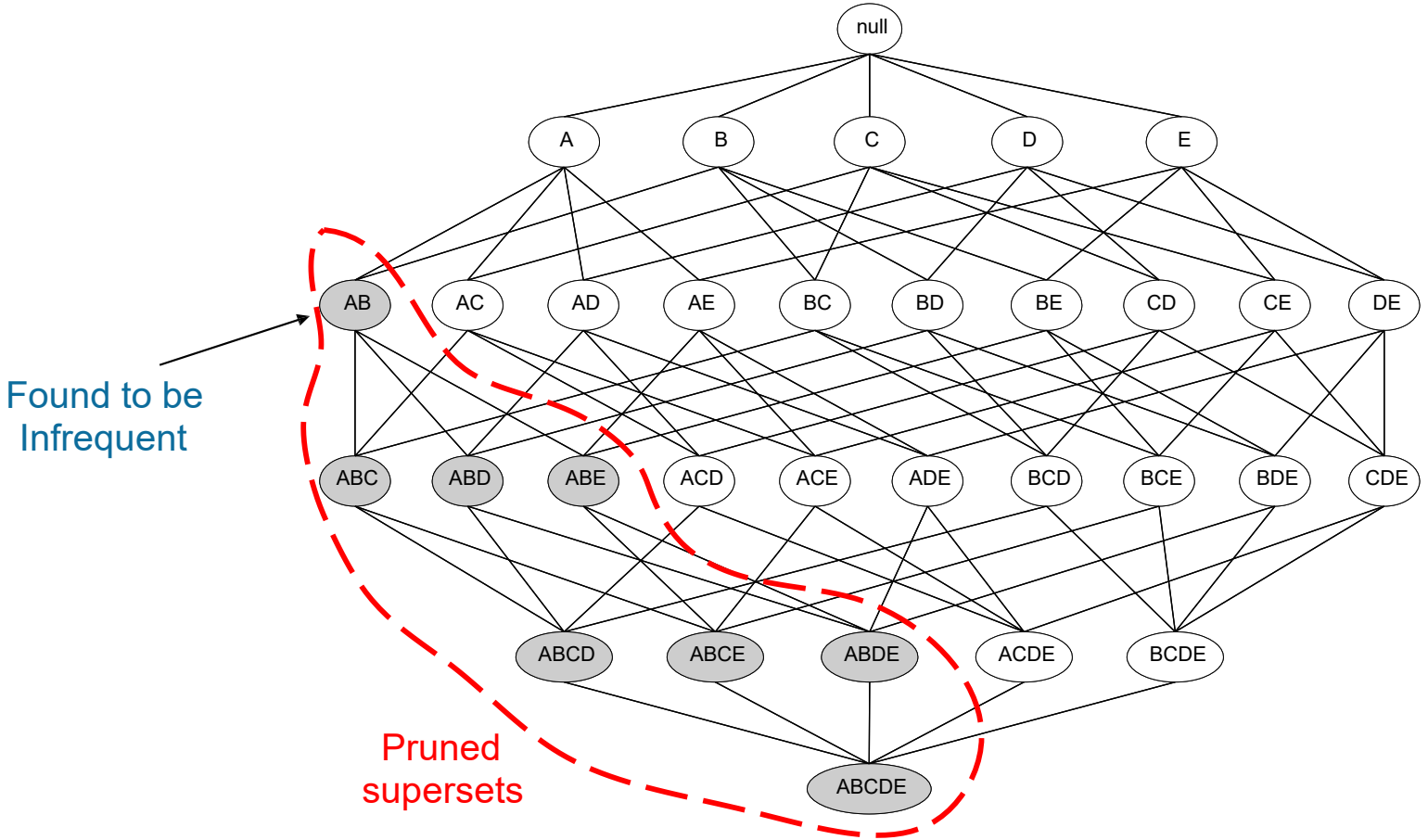
Apriori Principle (Agrawal & Srikant @VLDB'94):

- *If an itemset is frequent, then all of its subsets must also be frequent.*
- *Conversely, if an itemset is infrequent, then all of its supersets must also be infrequent.*

Thus we do not have to consider all possible itemsets. We can do pruning using Apriori principle.

- For example, if {beer, milk} is infrequent, then we do not need to consider {beer, milk, diaper} or {beer, milk, coke} and so on.

Illustration of Apriori Principle



Apriori Algorithm: Finding Freq. Itemsets by Confined Candidate Generation

F_k: frequent k-itemsets
C_k: candidate k-itemsets

Algorithm

- Let k=1
- Generate F_1 = {frequent 1-itemsets}
- Repeat until F_k is empty
  - **Candidate Generation**: Generate C_{k+1} from F_k
  - **Candidate Pruning**: Prune candidate itemsets in C_{k+1} containing subsets of length k that are infrequent
  - **Support Counting**: Count the support of each candidate in C_{k+1} by scanning the DB
  - **Candidate Elimination**: Eliminate candidates in C_{k+1} that are infrequent, leaving only those that are frequent in F_{k+1}

Apriori Algorithm Illustration

TID	Items
1	Bread, Milk
2	Beer, Bread, Diaper, Eggs
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Bread, Coke, Diaper, Milk



Items (1-itemsets)

C1 (candidate 1-itemsets)

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Minimum Support Count = 3 or 3/5 (Support)

Apriori Algorithm Illustration

Items (1-itemsets)
C1 (candidate 1-itemsets)

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1



Items (1-itemsets)
F1 (frequent 1-itemsets)

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Minimum Support Count= 3

Apriori Algorithm Illustration

Items (1-itemsets)

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Items (2-itemsets)
Candidate 2-itemsets

Itemset
{Bread,Milk}
{Bread, Beer}
{Bread,Diaper}
{Beer, Milk}
{Diaper, Milk}
{Beer,Diaper}

Pairs (2-itemsets) using joining

(No need to generate candidates involving Coke or Eggs)

Minimum Support Count= 3

Apriori Algorithm Illustration

Items (2-itemsets)
Candidate 2-itemsets

Itemset
{Bread,Milk}
{Bread, Beer }
{Bread,Diaper}
{Beer, Milk}
{Diaper, Milk}
{Beer,Diaper}



Items (2-itemsets)
Frequent 2-itemsets

Itemset	Count
{Bread,Milk}	3
{Beer, Bread}	2
{Bread,Diaper}	3
{Beer,Milk}	2
{Diaper,Milk}	3
{Beer,Diaper}	3

Minimum Support Count = 3

Apriori Algorithm Illustration

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Items (2-itemsets)

Frequent 2-itemsets

Itemset	Count
{Bread,Milk}	3
{Beer, Bread}	2
{Bread,Diaper}	3
{Beer,Milk}	2
{Diaper,Milk}	3
{Beer,Diaper}	3

Joining

Items (3-itemsets)

Candidate 3-itemsets

Itemset
{ Beer, Diaper, Milk}
{ Beer,Bread,Diaper}
{Bread, Diaper, Milk}
{ Beer, Bread, Milk}

Apriori Algorithm Illustration

Items (3-itemsets)
Candidate 3-itemsets

Itemset
{ Beer, Diaper, Milk}
{ Beer,Bread,Diaper}
{Bread, Diaper, Milk}
{ Beer, Bread, Milk}

Pruning

Items (3-itemsets)
Three 3-itemset are removed & pruned

Itemset	Count
{ Beer, Diaper, Milk}	2
{ Beer,Bread, Diaper}	2
{Bread, Diaper, Milk}	2
{Beer, Bread, Milk}	1

Itemset	Count
{Bread,Milk}	3
{Beer, Bread}	2
{Bread,Diaper}	3
{Beer,Milk}	2
{Diaper,Milk}	3
{Beer,Diaper}	3

Apriori Algorithm Illustration

Items (One Candidate 3-itemsets Left after Pruning)

NO Freq. 3-itemsets

Itemset	Count
{ Beer, Diaper, Milk}	2
{ Beer,Bread, Diaper}	2
{Bread, Diaper, Milk}	2
{Beer, Bread, Milk}	1



Apriori Stops

Minimum Support Count = 3

Apriori Algorithm Illustration

- If we use Brute-force method, how many matchings?

$$\binom{6}{1} + \binom{6}{2} + \binom{6}{3} = \mathbf{6 + 15 + 20 = 41}$$

- How many matchings with pruning in this Apriori algorithm example?

  6 + 6 + 1 = 13

- How many matchings without pruning in this Apriori algorithm example? (the worst case)

  6 + 6 + 4 = 16

Generating Association Rules from Freq. Itemsets

- Once the frequent itemsets from transactions in a database D have been found, it is straightforward to generate strong association rules from them (satisfy both minimum support and minimum confidence)

$$confidence(A \rightarrow B) = P(B|A) = \frac{support_count(A \cup B)}{support_count(A)}$$

- The conditional probability is expressed in terms of itemset support count, where $support_count(A \cup B)$ is the number of transactions containing the itemsets $A \cup B$, and $support_count(A)$ is the number of transactions containing the itemset A.

Generating Association Rules from Freq. Itemsets

- Based on this equation, association rules can be generated as follows:
  - For each **frequent itemset** I, generate all nonempty subsets of I
  - For every **nonempty subset** s of I, output the rule "$s \rightarrow (I - s)$" if $\frac{support_count(I)}{support_count(s)} \geq \min_conf$, where *min_conf* is the minimum confidence threshold.

Example: If {A,B,C,D} is a frequent itemset, candidate rules:

ABC → D, ABD →C, ACD →B, BCD →A

AB →CD, AC →BD, AD →BC, BC →AD, BD →AC, CD →AB

A →BCD, B →ACD, C →ABD, D →ABC

Note: If |I| = k, then there are 2^k-2 candidate association rules (without considering I → ∅ and ∅ → I)

Generating Association Rules from Freq. Itemsets

- If we have a frequent itemset X={I1, I2, I5}. What are the association rules that can be generated from X?
- The nonempty subsets of X are {I1, I2}, {I1, I5}, {I2, I5}, {I1}, {I2}, and {I5}
- The resulting association rules can be as follows:

{I1, I2} →{I5} (X - {I1, I2}) confidence = 2/4=0.5 or 50%

{I1, I5} → **{I2}** confidence =2/2=100%

{I2, I5} → **{I1}** confidence =2/2=100%

{I1} → **{I2, I5}** confidence =2/6 = 33%

{I2} → **{I1, I5}** confidence =2/7 = 29%

{I5} → **{I1, I2}** confidence = 2/2 = 100%

TID	*List of item_IDs*
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

If the minimum confidence is set to 70%, which rules are output as strong rules?

Rule Generation

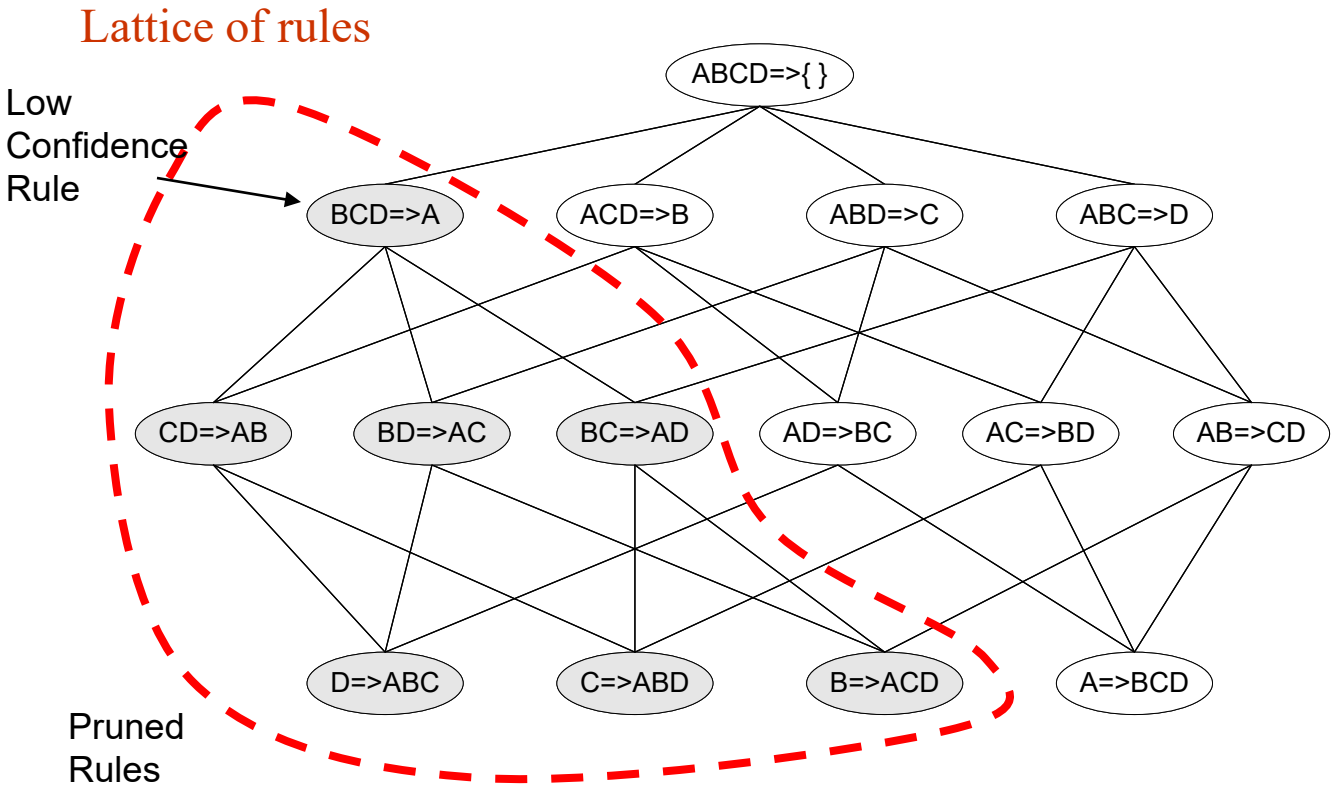
Confidence of rules generated from the same itemset has **an anti-monotone property**

- E.g., Suppose {A,B,C,D} is a frequent 4-itemset:

$$c(ABC \rightarrow D) \geq c(AB \rightarrow CD) \geq c(A \rightarrow BCD)$$

- Confidence is anti-monotone w.r.t. number of items on the RHS of the rule

Rule Generation for Apriori Algorithm



Mining Frequent Patterns I

- Basic Concepts
- Frequent Itemset Mining Methods
- Which Patterns Are Interesting?— Pattern Evaluation Methods
- Summary

Limitation of Thresholds

- A support–confidence framework (Most association rule mining algorithms).
  - Minimum support and confidence thresholds help weed out or exclude the exploration of a good number of uninteresting rules
  - Many of the rules generated are still not interesting to the users. Especially true when mining at low support thresholds, or mining for long patterns.
- This has been **a major bottleneck** for successful application of association rule mining.

"How can we tell which strong association rules are really interesting?"

A misleading "strong" association rule

- In an electronic device retailer data set, 10,000 transactions are analyzed, the data show that 6000 of the customer transactions included **computer games**, while 7500 included videos, and 4000 included both computer games and videos.
- Suppose **minimum support =30%** and **minimum confidence = 60%**. The following association rule is discovered:

Rule: Buys(X, "computer games")→buys(X, "videos")

[support = 40%, confidence = 66%]

- It **IS** a strong rule, but **this rule is misleading because the probability of purchasing videos is 75%, which is even larger than 66%**.

From Association Analysis to Correlation Analysis

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A correlation measure can be used to augment the support–confidence framework for association rules.

Assumption: The occurrence of itemset A is independent of the occurrence of itemset B, if $\mathrm{P}(\mathrm{A} \cup \mathrm{B}) = \mathrm{P}(\mathrm{A})\mathrm{P}(\mathrm{B})$. Otherwise, itemsets A and B are dependent and correlated events.

Lift is a simple correlation measure that is given as follows.

$$lift(A,B) = \frac{P(A \cup B)}{P(A)P(B)} = \frac{conf(A \rightarrow B)}{\sup(B)}$$

- If the lift is less than 1, *A* is *negatively correlated with B*.
- If the lift value is greater than 1, *A* and *B* are *positively correlated*.
- If the lift value is equal to 1, *A* and *B* are *independent* and there is no correlation between them.

Correlation Analysis Using Lift

In an electronic device retailer data set, 10,000 transactions are analyzed, the data show that 6000 of the customer transactions included computer games, while 7500 included videos, and 4000 included both computer games and videos.

- According to **the lift equation:**

$$lift(A,B) = \frac{P(A \cup B)}{P(A)P(B)} = \frac{conf(A \rightarrow B)}{\sup(B)}$$

- The corresponding lift(game, video) = P({game, video)/P({game})P({video}) = 0.4/(0.6*0.75) = 0.89 <1
- Thus, even the rule is strong according to the min_sup & min_conf, it is still not an interesting rule for decision making.

Summary

- Basic concepts:
  - Itemset & k-itemset
  - Support count/frequency
  - Frequent itemset
  - association rules, support-confident framework
  - Strong rule
- Apriori (Candidate generation & test)
  - Frequent Itemsets Generation
  - Rule Generation
- Which patterns are interesting?
  - Lift

Have a question?

